CH 3 - Random Variables

Luke Lu • 2026-10-21

3.1 Random Variables

> 📘 **Definition — Random variables**
>
> A **random variable** is a fucntion X that maps points in a smaple space to real numbers
>
> $$X : S \to \mathbb{R}$$
>
> A random variable can be discrete or continuous depending on whether the **range** is countable or uncountable

Example

Roll 3 fair 6-sided dice, $S = \{(1,1,1), ..., (6,6,6)\}$, 216 pairs

List a few possible random variables

$$X = \text{sum be the 3 dice, range} = \{3, ..., 18\},$$
$$Y = \text{max of 3 dice, range} = \{1, 2, 3, 4, 6\}$$
$$Z = \text{product of the 3 dice, range} = \{1, ..., 216\}$$

Continuous random variables could include anything measured in time, temperature, heightes, weights, etc

> 📘 **Definition — Probability Mass Function**
>
> The **PMF** of a discrete random variable $X$is
>
> $$\begin{aligned} f(x) &= P(X = x), \quad \forall x \in \text{range } X \\ &= \sum P(a_i) \quad \text{sum all the probability of the sample points that map to x} \end{aligned}$$

Examples

1. $X = \text{sum of 3 dice}$

   $$f(x) = \begin{cases} \frac{1}{216} \text{ if } x{=}3 \\ \frac{3}{216} \text{ if } x{=}4 \\ \dots \\ \frac{1}{216} \text{ if } x{=}18 \end{cases}$$

2. $W = 3 \text{ diigt number}$

   $$f(w) = \frac{1}{216}$$

3. $Y = \text{max of 3 dice}$

   $$f(y) = \begin{cases} \frac{1}{216} \text{ if } y{=}1 \\ \frac{7}{216} \text{ if } y{=}2 \\ \dots \end{cases}$$

Note: for any pmf, $f(x) > 0$ and $\sum\left(f_{x_i}\right) = 1$ because the random variable has to take on of the values

📘 **Definition — Probability Mass Function**

📘 **Definition — Cumulative Density Funciton**

3.2 Functions of Random Variables

In practice, we may be interested in some function of a random variable rather than the random variable itself.

i.e. total revenue based on units sold

If $Y = g(X)$, we may want the pmf of Y.

Example:

$X =$ number of 6 on 3 dice

Let $Y = \begin{cases} 0 \text{ if } x=0 \\ 1 \text{ if } x=1 \text{ or } 2 \\ 2 \text{ if } x=3 \end{cases}$, with range$\{0, 1, 2\}$

Y	0	1	2
$f_Y(y)$	$\frac{125}{216}$	$\frac{90}{216}$	$\frac{1}{216}$

$f_Y(0) = P(Y = 0) = P(X = 0) = f_X(0)$

$f_Y(1) = P(Y = 1) = P(X = 1 \text{ or } X = 2) = f_X(1) + f_X(2)$

$f_Y(2) = f_X(3)$

💡 **Tip** — Interchanging probability mass function variables

In general, if $Y = g(X)$ for a discrete random variable X

$$f_Y(y) = \sum_{\text{all } x \in \text{ range s.t. } g(x)=y} f_X(x)$$

We can view that as

$$f_Y(y) = P(Y = y) = P(X \text{ tales a value that } g(x) \to y) = \sum_{g(x)=y} P(X = x)$$

Example:

At a BINGO event, players pay 14 plus 2 per “Tonnie pot” bets(maximum bets be 2).

Let $X =$ the number of Toonie bets a player

X	0	1	2
$f_X(x)$	0.4	0.1	0.5

Let $Y =$ amount paid by random player, then $Y = 14 + 2X$

Y	14	16	18
$f_Y(y)$	0.4	0.1	0.5

Definition — Proposition

If $Y = g(X)$ is a **one-to-one** function on X with inverse $g^{-1}(Y) = X$ then

$$f_Y(y) = f_X(g^{-1}(y))$$

3.3 Expectation of Random Variables

Definition — Mean, Median, Mode

Mean is average

Median is the value where half are above and half are below

Mode is the most reoccurring data

1. Let $X =$ number of kids in a family,

   we have the sample data:

X	1	2	3	4	5+
frequency	28	81	26	5	10
proportion %	18.67	54	17.33	3.33	6.67

   Mean ≈ 2.25

   Median = Mode = 2

2. Let $X =$ number of courses

   we have the sample data:

X	1 or 2	3	4	5	6+
proportion %	1	1	12	72	13

   Median = Mode = 5

   Mean ≈ 5

We can rewrite the first example mean as $\sum_{i=1}^{5} x \cdot \text{proportion with value } x$

Definition — Expectation of Random Variable

The **Expected Value** of a discrete random variable

$$E[x] = \mu = \sum_{\text{all } x \in \text{ range}} x f(x)$$

Example:

Refer to the previous example.

According to StatCan, the pmf of $X =$ number of kids is

X	1	2	3	4
$f(x)$	0.44	0.36	0.17	0.03

$E[x] = 1.79$

We noticed that for every class selected, the average is always higher than the published result.

The one of the strongest reason is that the class selected is already complete while the published data family could be expanding.

That is whenever selecting a model, make sure that it is representative.

📘 **Definition — Expected Value of Function**

The **Expected Value** of a funciton of a discrete random variable X of $g(x)$ is

$$E[g(x)] = \sum_x g(x)f(x)$$

We can visualize as let $Y = g(y)$,

$$E[g(X)] = E[Y] = \sum_y yf_Y(y) = \sum_y \sum_x f_{X(x)} = \sum_y \sum_x yf_{X(x)} = \sum_y g(x)f_X(x)$$

Expectation is a linear operator.

For $a, b \in \mathbb{R}$ and discrete random variable X

$$\begin{aligned} E[aX + b] &= aE[x] + b = \sum_x (ax + b)f(x) \\ &= \sum_x axf(x) + \sum_x bf(x) = a\sum_x xf(x) + b\sum_x f(x) \\ &= aE(x) + b \end{aligned}$$

⚠️ **Warning — Common Mistakes**

$$E[g(x)] \neq g(E(x))$$

Example: Roulette

38 slots with 18 Black, 18 Red, and 2 Green

You can bet on a
- single number pays 35:1
- two numbers pays 17:1
- four numbers pays 8:1
- column of 1/3 numbers pays 2:1
- colour, parity, 1/2 number pays 1:1

Strategy 1: 100 on single number

if $X =$ number rolled, $g_1(X) = \begin{cases} 3600 \text{ with probability } \frac{1}{38} \\ 0 \text{ with probability } \frac{37}{38} \end{cases}$

$E[g_1(X)] = 3600 \cdot \frac{1}{38} + 0 \cdot \frac{37}{38} = 94.74$

Definition — Variance

We define the **variance**, $\text{Var}(X) = \sigma^2 = E[(X-\mu)^2] = E[(X - E[X]^2)]$

In practice, we can evaluate the variance in **DISCRETE** random variable by

$$\begin{aligned} \text{Var}(X) &= E[(X-\mu)^2] = E[X^2] - 2\mu E[X] + \mu^2 \\ &= E[X^2] - \mu^2 = E[X^2] - E[X]^2 \end{aligned}$$

Definition — Standard Deviation

We define the **standard deviation**,

$$\sigma(X) = \sqrt{\text{Var(X)}}$$

Example:

Roulette

$X = \begin{cases} 36 \text{ with } \frac{1}{38} \\ 0 \text{ with } \frac{37}{38} \end{cases}$

$\text{Var}(X) = E[(X-\mu)^2] = E[X^2] - E[X]^2 = 34.105 - 0.94737^2 \approx 33.207\$^2$

$\sigma(X) = \sqrt{\text{Var}(X)} \approx 5.76\$$

Let $Y = \begin{cases} 2 \text{ with } \frac{18}{38} \\ 0 \text{ with } \frac{20}{38} \end{cases}$

$\text{Var}(Y) = E[Y^2] - E[Y]^2 = 2^2 \cdot \frac{18}{38} - 0.94737^2 \approx 0.99723$

$\sigma \approx 0.99861$

Let $Z = \begin{cases} 1 \text{ with } \frac{36}{38} \\ 0 \text{ with } \frac{2}{28} \end{cases}$

$\text{Var}(Z) = E[Z^2] - E[Z]^2 = \frac{36}{38} - 0.94737^2 \approx 0.04986$

$\sigma \approx 0.2233$

Warning — Variance, Standard deviation

Variance is NOT a linear operator.

$$\begin{aligned} \text{Var}(aX+b) &= E[((aX+b) - E[aX+b])^2] \\ &= E[(aX + b - aE(X) - b)^2] = E[(aX - aE([X])^2)] \\ &= a^2 E[(X - E(X))^2] = a^2 \ \text{Var}(X) \end{aligned}$$

Standard deviation

$\sigma(aX+b) = |a|\sigma(X)$

3.4 Moment Generating Functions

Definition — Moment

The k-th moment of a random variable X is $E[X^k]$, for $k = 1, 2, 3, ...$

First moment is $E[X]$, is the mean

Second moment is $E[X^2]$, needed for computation of variance

Third moment is $E[X^3]$, needed for computation of skewness

Fourth moment is $E[X^4]$, needed for computation of kurtosis.

Definition — Moment Generating Function

The MGF of a random variable defined as X is $M_X(t) = E[e^{tx}]$ where it exists in some region around $t = 0$

- It uniquely determines a distribution.
- Can easily prove relationships between random variables.
- Determine the moments more quickly then using the $f(x)$ or $F(x)$

In general

$$E[X^k] = M_X^{(k)}(0) = \frac{d}{\mathrm{d}t} M_X(t)\Big|_{t=0}$$

Properties of MGF

- $M_{X(0)} = 1, \ \forall X$
- $M'_X(t) = \frac{d}{\mathrm{d}t} E[e^{tx}] = \frac{d}{\mathrm{d}t} = E\left[1 + X + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + ...\right] = E\left[0 + X + tX^2 + \frac{1}{2}t^2X^3 + ...\right]$
- $M'_X(0) = E[X]$
- $M''_X(t) = E[0 + 0 + X^2 + tX^3 + ...]$
- $M''_X(t) = E[X^2]$

Example:

Suppose we roll a fair six-sided die until to get 6. Let $X = \text{number of non-six rolled}$. Use MGF to find the mean.

$f(x) = \left(\frac{5}{6}\right)^x \left(\frac{1}{6}\right), x = 0, 1, 2, 3, 4, ...$

$M_X(t) = E[e^{tx}] = \sum_{X=0}^{\infty} e^{tx} \left(\frac{5}{6}\right)^x \left(\frac{1}{6}\right) = \frac{1}{6} \sum_{X=0}^{\infty} \left(\frac{5}{6}e^t\right)^x = \frac{1}{6} \frac{1}{1 - \frac{5}{6}e^t}$, where $|\frac{5}{6}e^t| < 1$

$M'_X(0) = (6 - 5e^t)^{-1}\Big|_{t=0} = 5e^t(6 - 5e^t)^{-2}\Big|_{t=0} = 5$

Let $Y = aX + b$, $M_Y(t) = E[e^{taX+bt}] = E[e^{taX}e^{bt}] = e^{bt}E[e^{taX}] = e^{bt}M_X(at)$